# C-Band 6×6 Dual-Polarised Antenna Array

Results summary and project status

| | |
|---|---|
| Document | TM-RF-15-002, Revision A |
| Date | 30 October 2026 |
| Report type | Results and status report |
| Frequency band | 4.0–7.0 GHz (C-band) |
| Prepared by | Telmart RF & Microwave Engineering |
| Status | In progress: full-array verification and element development |

Confidential. This document presents results only. Methods, analysis models and design concepts are proprietary to Telmart and are not disclosed. Not to be reproduced or passed to third parties without the written consent of Telmart.

# 1 Summary of results

**5.35–6.26 GHz**
Present element matched (VSWR ≤ 3): 16 % bandwidth

**54 %**
Bandwidth required for 4–7 GHz

**8.7–9.8 dBi**
Element gain (accepted power), 5–6.5 GHz

**> 20 dB**
Isolation between V and H inputs over most of the band (array)

**Main result.** The poor input match reported on the array is caused by the radiating element itself, whose matched bandwidth is about 16 %, well below the 54 % needed for 4–7 GHz. The radiator works correctly inside its band, which is why the gain is as expected while the VSWR is poor.

**Consequence.** A change to the feed network alone cannot bring the VSWR below 3:1 across 4–7 GHz. A new wideband radiating element is required; its development has started (section 5).

**Measured panel.** Bench measurements confirm the input-match behaviour and show a gain of 20.4 dBi peak, about 3 dB below the 23 dBi rating, with a strong drop above 6 GHz (section 6).

# 2 Design reviewed

| Item | Value |
|---|---|
| Array | 6 × 6 dual-linear-polarised stacked patches, 276.86 × 276.86 mm, pitch ≈ 44 mm |
| Inputs | Input 1: V-polarisation; Input 2: H-polarisation (one 36-way feed network each) |
| Stack-up | Ground / RT5880 0.8 mm / driven patches and feeds / 4 mm air / FR4 0.4 mm / parasitic patches |
| Reported symptom | Gain as expected; VSWR above 3:1 over most of 4–7 GHz |

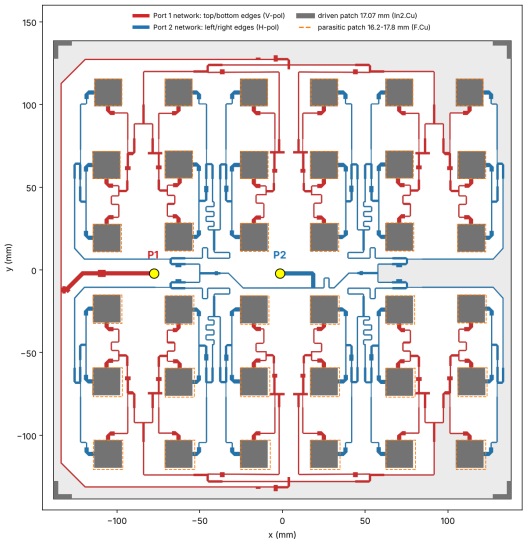



**Figure 1.** Array layout reviewed (client design). Red: input 1 feed network; blue: input 2 feed network.

# 3 Element results (present design)

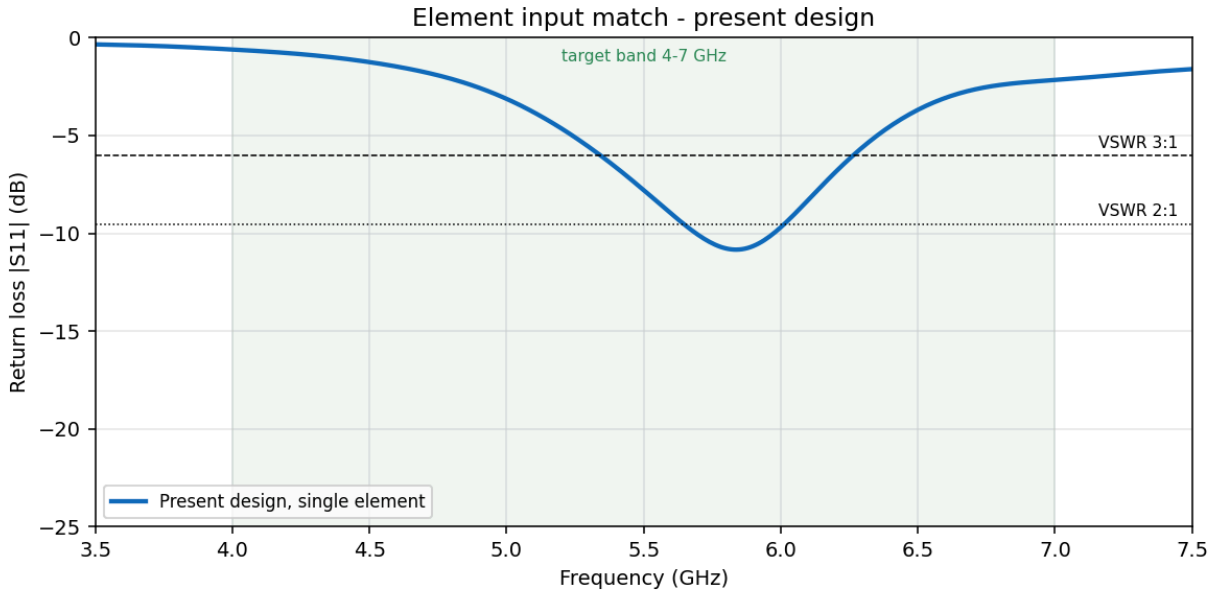



**Figure 2.** Input match of one element of the present design. Dashed line: VSWR 3:1; dotted: VSWR 2:1; shaded: 4–7 GHz target band.

| Frequency (GHz) | Return loss \|S11\| (dB) | VSWR |
|---|---|---|
| 4.0 | -0.61 | 28.6 |
| 4.5 | -1.25 | 13.9 |
| 5.0 | -3.13 | 5.6 |
| 5.5 | -7.79 | 2.4 |
| 6.0 | -9.69 | 2.0 |
| 6.5 | -3.69 | 4.8 |
| 7.0 | -2.17 | 8.1 |

| Frequency (GHz) | Gain, accepted power (dBi) | Realised gain incl. mismatch (dBi) |
|---|---|---|
| 5.0 | 8.70 | 5.81 |
| 5.5 | 9.25 | 8.46 |
| 6.0 | 9.74 | 9.25 |
| 6.5 | 9.79 | 7.36 |

The element is matched only between 5.35 and 6.26 GHz. Outside this window the realised gain falls because of the mismatch, while the gain referred to accepted power stays at 8.7–9.8 dBi.

# 4 Complete array results (preliminary)

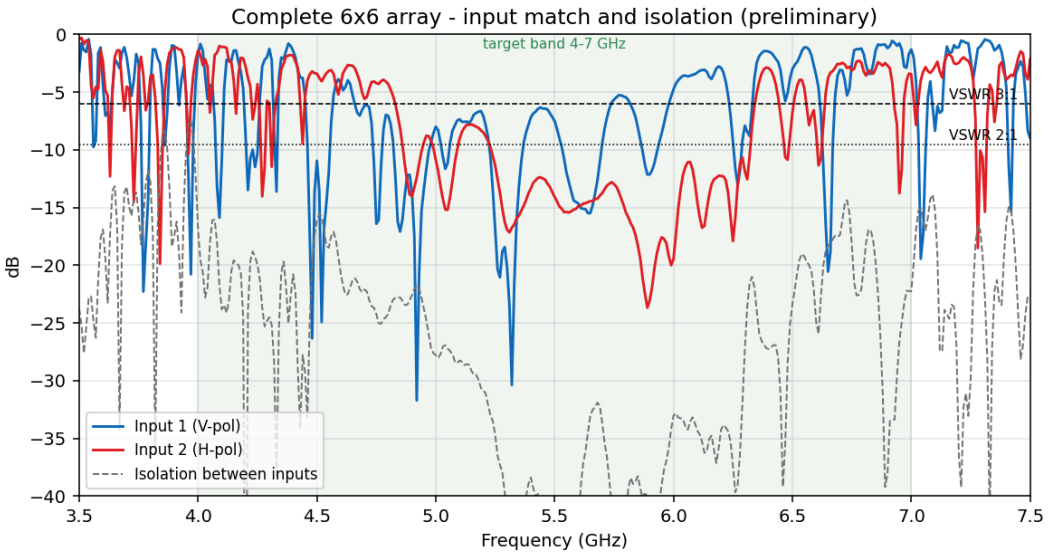



**Figure 3.** Complete 6×6 array: input match of both inputs and isolation between them. Preliminary: final values follow in the next revision.

- Both inputs are best in the middle of the band (about 5.0–6.2 GHz).
- Below about 4.8 GHz and above about 6.3 GHz both inputs are mostly at VSWR 3:1 or worse, consistent with the element result.
- Isolation between the V and H inputs is better than 20 dB over most of the band.
- The array result confirms that the element bandwidth is the limiting factor.

# 5 Element development (in progress)

A new wideband element is being developed to meet VSWR < 3 (target < 2) over the full 4–7 GHz band on both polarisations. Iteration 1 has been completed; it is matched from 4.95 to 5.75 GHz and has defined the design direction for iteration 2.

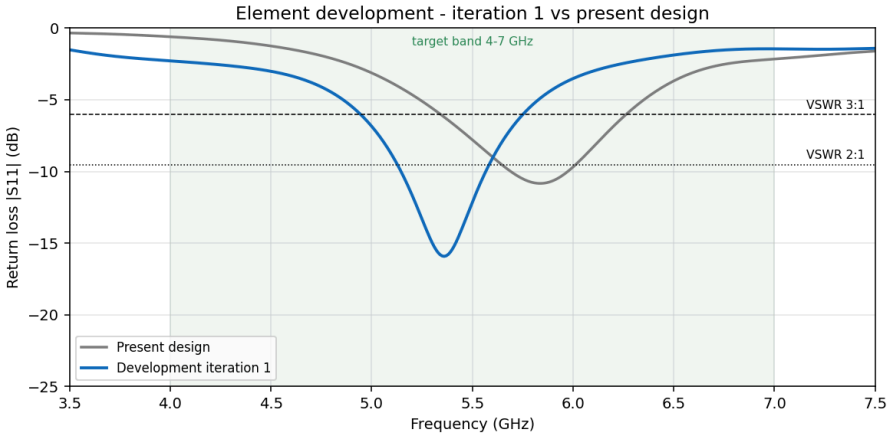



**Figure 4.** Element development: iteration 1 compared with the present design.

# 6 Measured results of the built panel

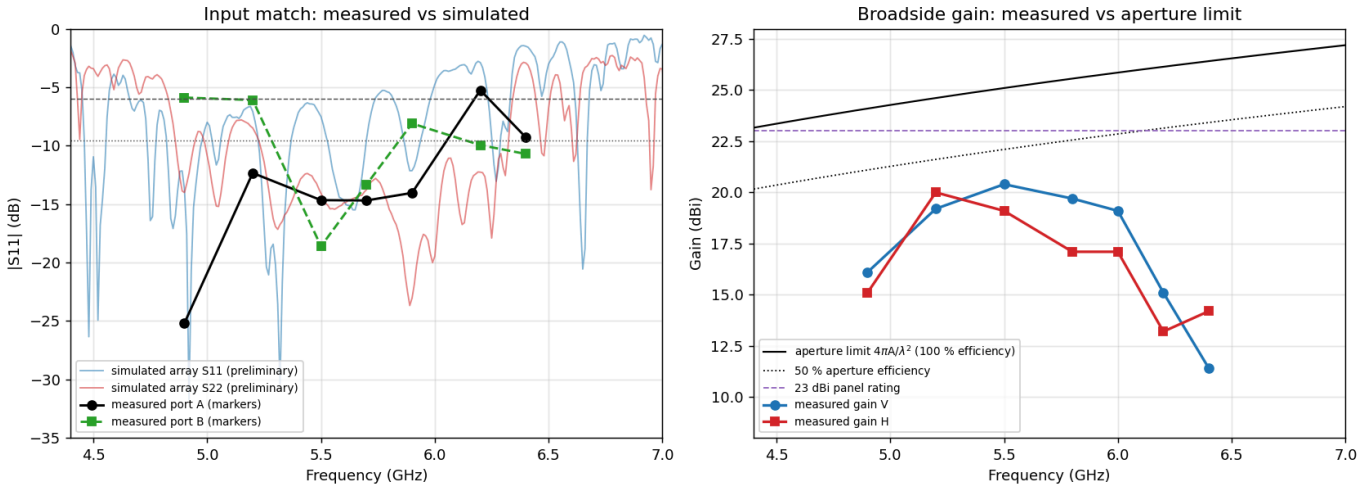



**Figure 5.** Built panel: measured input match (markers) against the predicted array response (left), and measured gain against the aperture limit and the 23 dBi rating (right).

| f (GHz) | S11 port A (dB) | S11 port B (dB) | Gain V (dBi) | Gain H (dBi) | Aperture limit (dBi) | Aperture efficiency V / H |
|---|---|---|---|---|---|---|
| 4.9 | -25.2 | -5.9 | 16.1 | 15.1 | 24.1 | 16 % / 13 % |
| 5.2 | -12.3 | -6.1 | 19.2 | 20.0 | 24.6 | 29 % / 35 % |
| 5.5 | -14.7 | -18.6 | 20.4 | 19.1 | 25.1 | 34 % / 25 % |
| 5.8 | -14.7* | -13.3* | 19.7 | 17.1 | 25.6 | 26 % / 14 % |
| 6.0 | -14.0* | -8.1* | 19.1 | 17.1 | 25.9 | 21 % / 13 % |
| 6.2 | -5.3 | -9.9 | 15.1 | 13.2 | 26.1 | 8 % / 5 % |
| 6.4 | -9.3 | -10.7 | 11.4 | 14.2 | 26.4 | 3 % / 6 % |

* nearest S11 marker (5.7 / 5.9 GHz). Gain by substitution against a reference horn at ~5 m. Port A/B to V/H mapping and radome condition during the gain test to be confirmed.

- The measured input match agrees with the predicted array response: best in the middle of the band, poorer at the band edges.
- Measured peak gain is 20.4 dBi (V) and 20.0 dBi (H), about 3 dB below the 23 dBi rating.
- Gain drops to 11–15 dBi at 6.2–6.4 GHz. The input mismatch accounts for at most about 1.5 dB of this; the cause of the remaining drop is under investigation.

# 7 Project status

| # | Work package | Status | Result / next output |
|---|---|---|---|
| 1 | Design data review (layout and stack-up) | ● Completed | Design decoded, both feed networks traced |
| 2 | Element performance analysis, present design | ● Completed | Root cause identified: element bandwidth ~16 % |
| 3 | Feed-network correction assessment | ● Completed | Feed-network change alone is not sufficient |
| 3b | Bench measurement of the built panel (input match, gain) | ● Completed | Input match confirmed; gain shortfall identified (section 6) |
| 3c | Gain shortfall investigation | ◐ In progress | Cause of the ~3 dB shortfall and the high-band drop |
| 4 | Complete-array verification, present design | ◐ In progress | Preliminary results issued (section 4); final values in Rev. B |
| 5 | Wideband element development | ◐ In progress | Iteration 1 completed; iteration 2 is the next step |
| 6 | Array integration of the new element | ○ Planned | Updated array layout and simulated performance |
| 7 | Prototype and measurement | ○ Planned | Measured input match, isolation and patterns |

● completed ◐ in progress ○ planned

# 8 Actions requested from the client

- Confirm the required specification: VSWR limit (3:1 or 2:1), band edges, gain and isolation targets.
- Confirm any mechanical constraints for the new element (total height, materials, array size and pitch).
- Confirm which panel port is V and which is H, and whether the gain was measured with the radome fitted.
- If possible, repeat the gain measurement with and without the radome, and provide pattern cuts at 5.5 and 6.4 GHz.